

Simple linear regression

- The sample: $\{(x_i, y_i)\}_{i=1}^n$

independent
variable

dependent
variable

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon$$

$$\varepsilon \sim N(0, \sigma^2)$$

random deviation or random error

Multiple linear regression with k independent variables. $k > 1$

- The sample: $\{(x_{1i}, x_{2i}, \dots, x_{ki}, y_i)\}_{i=1}^n$



A red arrow points from the x terms in the sample notation to the 'independent variables' box. A blue arrow points from the y term in the sample notation to the 'dependent variable' box.

independent
variables

dependent
variable

$$Y = \underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k}_{\text{True regression line}} + \varepsilon \quad \varepsilon \sim N(0, \sigma^2)$$

True regression line

random deviation or random error

Summary of Multiple LR

The sample:

$$\{(x_{1i}, x_{2i}, \dots, x_{ki}, y_i)\}_{i=1}^n$$

$$Y = \underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k}_{\text{True regression line}} + \varepsilon \quad \varepsilon \sim N(0, \sigma^2)$$

- Significance of LR model
 - $H_0: \beta_i = 0$ for all i
 $H_a: \beta_i \neq 0$ for at least one i
F-test for the whole model
 - t-tests for individual independent variables. $H_0: \beta_i = 0$
 $H_a: \beta_i \neq 0$
- Percentage of variability explained by the LR model
 - R^2 : no reference to df
 - Adjusted R^2 : adjusted according to df so as to minimize MSE

Subjects

- Problems: Confounding and Collinearity
- Model selection:
 - Principle of Parsimony: “A model should contain the smallest number of variables necessary to fit the data.”

$$H_0: \beta_{k+1} = \cdots = \beta_p = 0$$

- Best subsets:
 - Full vs reduced models
 - Stepwise regression (partial F-tests): Forward selection & Backward elimination
- Criteria: coefficient of determination, R^2 , adjusted R^2 , Mallows' C_p , MSE

$$f = \frac{(\text{SSE}_{\text{reduced}} - \text{SSE}_{\text{full}})/(p - k)}{\text{SSE}_{\text{full}}/(n - p - 1)} \sim F_{p-k, n-(p+1)}$$

Stepwise Regression

- This is the most widely use model selection technique.
- Its main advantage over best subsets regression is that it is less computationally intensive, so it can be used in situations where there are a very large number of candidate independent variables and too many possible subsets for every one of them to be examined.
- The version of stepwise regression that we will describe is based on the P -values of the t statistics for the independent variables. An equivalent version is based on the F statistic (which is the square of the t statistic).

Stepwise Regression

- Before running the algorithm, the user chooses two threshold P -values, α_{in} and α_{out} , with $\alpha_{\text{in}} \leq \alpha_{\text{out}}$.
- The stepwise regression procedure begins with a step called a **forward selection** step, in which the independent variables with smallest P -value is selected, provided that $P < \alpha_{\text{in}}$.
- This variable is entered in the model, creating a model with a single independent variable.

More on Stepwise Regression ¹

- In the next step, the remaining variables are examined one at a time as candidates for the second variable in the model. The one with the smallest P -value is added to the model, again provided that $P < \alpha_{\text{in}}$.
- Now, it is possible that adding the second variables to the model increased the P -value of the first variable. In the next step, called a **backward elimination** step, the first variable is dropped from the model if its P -value has grown to exceed the value α_{out} .

More on Stepwise Regression

- The algorithm continues by alternating forward selection steps with backward eliminations steps.
- The algorithm terminates when no variables meet the criteria for being added to or dropped from the model.

Class Exercise: Mobile Ad Hoc Computer Networks

- The resulting model is not in the best subsets.
- The interaction term is in the model, but the individual factors are not.

Alpha-to-Enter = 0.15 Alpha-to-Remove = 0.15

Response is Overhead on 7 predictors, with N = 25

Step	1	2	3	4	5
Constant	452.2	437.3	410.7	388.4	338.5
Speed*Pause	0.0470	0.0355	0.0355	0.0304	0.0146
T-Value	8.03	6.00	6.96	5.21	3.52
P-Value	0.000	0.000	0.000	0.000	0.002
Pause		0.74	3.02	3.75	3.24
T-Value		3.40	3.78	4.20	6.25
P-Value		0.003	0.001	0.000	0.000
Pause^2			-0.0380	-0.0442	-0.0256
T-Value			-2.94	-3.39	-3.19
P-Value			0.008	0.003	0.005
LCR				0.69	6.97
T-Value				1.62	6.95
P-Value				0.121	0.000
LCR^2					-0.139
T-Value					-6.46
P-Value					0.000
S	15.2	12.6	10.8	10.4	5.99
R-Sq	73.70	82.74	87.77	89.19	96.62
R-Sq(adj)	72.55	81.18	86.02	87.02	95.73
Mallows C-p	140.6	87.0	58.1	51.4	7.8

Class Exercise: Mobile Ad Hoc Computer Networks

- Dependent variable: Data overhead (# bytes of information to be transmitted along the message to get them to the right places)
- Independent variables: Speed, Pause time, Link change rate

Best Subsets Regression
Response is Overhead

Vars	R-Sq	Adj. R-Sq	C-p	s	S p e e d	P a u s e	L C R	S p e e d ^ 2	P a u s e ^ 2	s p e e d * p a u s e	L C R ^ 2
1	73.7	72.5	140.6	15.171						X	
1	54.5	52.6	258.3	19.946		X					
2	82.7	81.2	87.0	12.564		X				X	
2	82.2	80.6	90.3	12.755	X	X					
3	92.9	91.9	26.5	8.2340		X	X				X
3	89.6	88.1	46.9	9.9870	X		X	X			
4	95.9	95.0	10.3	6.4430	X	X	X				X
4	95.4	94.5	13.2	6.7991		X	X	X			X
5	96.8	95.9	6.7	5.8315	X	X	X		X		X
5	96.7	95.8	7.2	5.9074	X	X		X		X	X
6	97.1	96.2	6.6	5.6651	X	X	X	X	X		X
6	97.0	96.0	7.6	5.8164	X	X	X	X		X	X
7	97.2	96.1	8.0	5.7234	X	X	X	X	X	X	X

Notes on Model Selection

- When there is little or no physical theory to rely on, many different models will fit the data about equally well.
- The methods for choosing a model involve statistics, whose values depend on the data. Therefore, if the experiment is repeated, these statistics will come out differently, and different models may appear to be “best.”
- Some or all of the independent variables in a selected model may not really be related to the dependent variable. Whenever possible, experiments should be repeated to test these apparent relationships.
- Model selection is an art, not a science.

Simulation experiment

- $\{(x_{1i}, x_{2i}, \dots, x_{20}, y_i)\}_{i=1}^{30}$: all variables are generated independently as normal (0,1) variables

Stepwise Regression: Y versus X1, X2, ...

Alpha-to-Enter: 0.15 Alpha-to-Remove: 0.15

Response is Y on 20 predictors, with N = 30

Stepwise Regression

Step	1	2	3	4	5	6
Constant	0.14173	0.11689	0.12016	0.13756	0.09070	0.03589
X15	-0.38	-0.38	-0.28	-0.32	-0.28	-0.30
T-Value	-2.08	-2.19	-1.60	-1.87	-1.69	-1.89
P-Value	0.047	0.037	0.122	0.073	0.105	0.071
X6		0.39	0.55	0.57	0.57	0.52
T-Value		2.04	2.76	2.99	3.15	2.87
P-Value		0.051	0.010	0.006	0.004	0.009
X16			-0.43	-0.43	-0.55	-0.73
T-Value			-1.98	-2.06	-2.60	-3.07
P-Value			0.058	0.050	0.016	0.005
X12				0.33	0.42	0.49
T-Value				1.79	2.29	2.66
P-Value				0.086	0.031	0.014
X3					-0.42	-0.52
T-Value					-1.83	-2.23
P-Value					0.080	0.035
X17						0.35
T-Value						1.53
P-Value						0.140
S	1.15	1.09	1.04	0.998	0.954	0.928
R-Sq	13.33	24.92	34.75	42.15	49.23	53.91
R-Sq(adj)	10.24	19.36	27.22	32.90	38.66	41.89
Mallows C-p	5.5	3.3	1.7	1.0	0.4	0.7

Best Subsets

Vars	R-Sq	R-Sq(adj)	Mallows C-p	S																				
					1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0
1	13.3	10.2	5.5	1.1539															X					
2	28.3	23.0	2.0	1.0685						X										X				
3	34.8	27.2	1.7	1.0390						X									X	X				
4	43.2	34.1	0.6	0.98851			X			X						X				X				
5	49.2	38.7	0.4	0.95391			X			X						X			X	X				
6	53.9	41.9	0.7	0.92844			X			X						X			X	X	X			
7	57.7	44.3	1.3	0.90899	X	X				X						X			X	X			X	
8	61.2	46.4	2.1	0.89168			X			X						X	X		X	X	X		X	
9	65.0	49.3	2.7	0.86747			X			X	X					X	X		X	X	X		X	
10	67.6	50.5	3.8	0.85680			X			X	X					X	X		X	X	X	X	X	
11	69.2	50.4	5.2	0.85803	X	X	X			X	X				X	X			X	X	X		X	
12	71.3	51.0	6.4	0.85267	X	X	X			X	X				X	X		X	X	X	X		X	
13	72.4	49.9	8.0	0.86165	X	X	X			X	X			X	X	X		X	X	X	X		X	
14	73.0	47.8	9.8	0.87965	X	X	X			X	X	X		X	X	X		X	X	X	X		X	
15	74.2	46.5	11.4	0.89122	X	X	X			X	X	X		X	X	X		X	X	X	X	X	X	
16	74.5	43.1	13.3	0.91886	X	X	X		X	X	X	X		X	X	X		X	X	X	X	X	X	
17	74.8	39.2	15.1	0.94985	X	X	X		X	X	X	X		X	X	X	X	X	X	X	X	X	X	
18	75.1	34.2	17.1	0.98777	X	X	X		X	X	X	X		X	X	X	X	X	X	X	X	X	X	X
19	75.1	27.9	19.0	1.0344	X	X	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
20	75.2	20.1	21.0	1.0886	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

All relations are by chance!

Cross validation

There are advanced methods of variable selection that are somewhat more effective in constructing useful models. Dividing the data into two pieces, then using one (the training set) to fit the model and the other (the test set) to evaluate it, can be helpful. A method known as **cross-validation** employs a more advanced version of this idea. A procedure called the **lasso**, which is one of a class of methods known as shrinkage methods, has been shown to be useful in some situations. Finally, if it is possible to repeat the experiment to collect more data, the model constructed from the original data can be fit to the new data. If it fits well, the evidence of a real relationship becomes more convincing. More information can be found in James, Witten, et al., (2021).

Summary

When selecting a regression model, keep the following in mind:

- When there is little or no physical theory to rely on, many different models will fit the data about equally well.
- The methods for choosing a model involve statistics (R^2 , the F statistic, C_p), whose values depend on the data. Therefore if the experiment is repeated, these statistics will come out differently, and different models may appear to be “best.”
- Some or all of the independent variables in a selected model may not really be related to the dependent variable. Whenever possible, experiments should be repeated to test these apparent relationships.
- Model selection is an art, not a science.

How to choose the best regression model?

- Different criteria

- Higher adjusted coefficient of multiple determination:

$$r_a^2 = 1 - \frac{SSE/(n - k)}{SST/(n - 1)}$$

- Equivalently lower MSE
- Lower Mallows' C_p

Problems in multiple regression

1. Multicollinearity occurs when two or more of the regressors are “highly” correlated. Then the estimates of the corresponding β 's have high variances.

To detect multicollinearity:

1. Inspection of sample correlations between x_j and x_k for all j, k . If sample correlation $|r_{jk}| \approx 1$: suspect multicollinearity between x_j & x_k if more than 2 variables are correlated. this may not work.
2. Let x_j be one variable that is suspected to have high correlation with one or more of the other x_k .

Regression of x_j on x_k : If R^2 is “high” multicollinearity.

3. F-test for regression is significant, but individual tests on β_i are not. Then suspect multicollinearity.
2. Influential observations in regression (problem for simple regression as well).
3. Autocorrelation (simple regression as well).

Verifying the assumptions: Diagnostic Plots

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + \varepsilon_i$$
$$\varepsilon_i \sim N(0, \sigma^2)$$

Assumptions on ε_i :

- | | | |
|---|-----------------------|--|
| 1) e_i^* (or e_i) versus x_i | $\longleftrightarrow$ | 1) ε_i 's (so $E(\varepsilon_i)$ and $V(\varepsilon_i)$) do not depend on x_i . |
| 2) e_i^* (or e_i) versus y_i | $\longleftrightarrow$ | 2) ε_i 's (so $E(\varepsilon_i)$ and $V(\varepsilon_i)$) do not depend on y_i . |
| 3) normal probability plot of e_i^* (or e_i) | $\longleftrightarrow$ | 3) ε_i 's are normal. |
| 4) e_i^* (or e_i) versus i | $\longleftrightarrow$ | 4) ε_i 's are independent over time. i.e., do not depend on i . |

Verifying the assumptions: Diagnostic Plots

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + \varepsilon_i$$
$$\varepsilon_i \sim N(0, \sigma^2)$$

5) $\hat{y}_i$ versus y_i



5) close to a 45° line??

6) y_i versus x_i



6) an observation with
large influence

Example

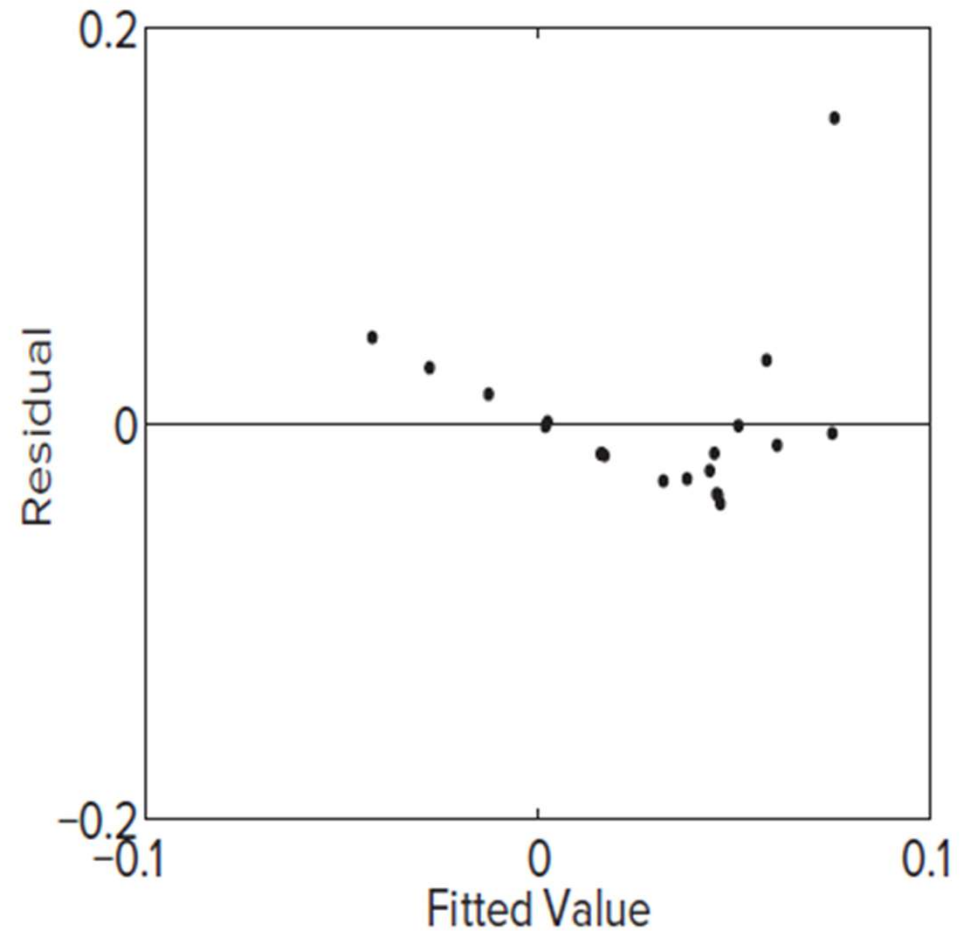
The article “Permanent Deformation Characterization of Subgrade Soils from RLT Test” (A. Puppala, L. Mohammad, et al., *Journal of Materials in Civil Engineering*, 1999:274–282) presents measurements of plastic strains (in percent) on soils at various confining and deviatoric stresses in kPa. The following table presents the plastic strains (y), the confining stress (x_1), and the deviatoric stress (x_2) for tests on a sandy soil.

y	x_1	x_2
0.01	21	21
0.02	21	35
0.05	21	52.5
0.09	21	70
0.003	35	35
0.006	35	70
0.05	35	105
0.23	35	140
0.003	70	35
0.0008	70	70
0.01	70	140
0.07	70	210
0.002	105	70
0.0003	105	105
0.0009	105	140
0.01	105	210
0.001	140	70
0.0003	140	105
0.0005	140	210
0.03	140	280

Model 1

Predictor	Coef	StDev	<i>T</i>	<i>P</i>
Constant	0.046975	0.021292	2.2062	0.041
x_1	-0.00084581	0.00028086	-3.0115	0.008
x_2	0.00041568	0.00017662	2.3536	0.031

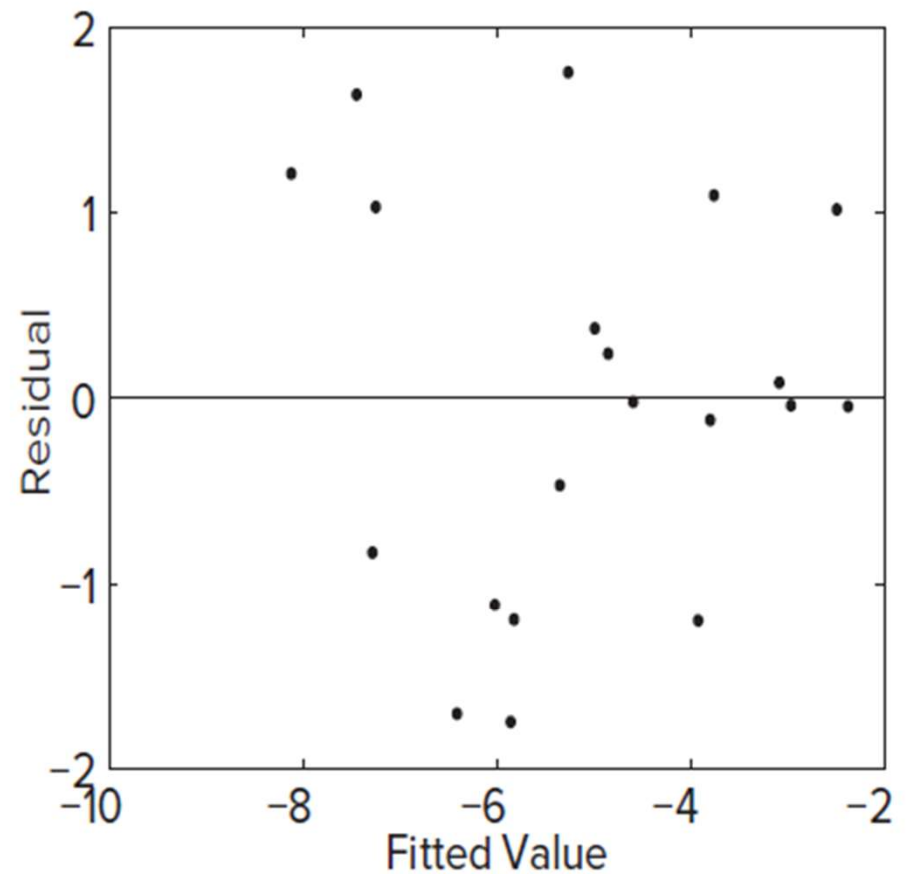
$$y = 0.046975 - 0.00084581x_1 + 0.00041568x_2.$$



Model 2

Predictor	Coef	StDev	<i>T</i>	<i>P</i>
Constant	-1.8898	1.761	-1.0732	0.298
$\ln x_1$	-3.0315	0.45905	-6.6039	0.000
$\ln x_2$	2.06	0.46648	4.416	0.000

$$\ln y = -1.8898 - 3.0315x_1 + 2.0600x_2.$$



Questions

- c. Use the more appropriate of the models from (a) and (b) to predict y when $x_1 = 50$ and $x_2 = 100$.

$$\ln \hat{y} = -1.8898 - 3.0315 \ln 50 + 2.0600 \ln 100 = -4.2624$$

$$\hat{y} = e^{-4.2624} = 0.01409.$$

Questions

d. Is the model you used in part (c) improved by including an interaction term? Explain.

Predictor	Coef	StDev	<i>T</i>	<i>P</i>
Constant	−5.6416	10.491	−0.53777	0.598
$\ln x_1$	−2.0919	2.6307	−0.79518	0.438
$\ln x_2$	2.9432	2.4793	1.1871	0.253
$\ln x_1 \ln x_2$	−0.2174	0.5988	−0.36306	0.721